

Weekly Progress Report - Week 03

Name: Shivani

Domain: Core Java Programming

Project: Banking Information System

Date of Submission: 15-07-2026

Week Ending: 03

I. Overview

During the third week of my Core Java internship, I focused on implementing the core modules of the Banking Information System. I applied the Java concepts learned during the previous weeks to develop practical project features. My work concentrated on user registration, account creation, account management, and improving the overall structure of the application. I followed object-oriented programming principles to design reusable classes and methods. I also spent time understanding how real banking applications organize customer information, account details, and transactions. Daily coding practice improved my logical thinking, debugging ability, and confidence in Java programming.

II. Objectives

The objectives for this week were to design reusable Java classes, implement user registration, create account management functionality, validate user input, improve modular programming, practice exception handling, and understand project architecture. These objectives helped me move from theoretical learning to practical software development.

III. Daily Work Progress

Day 1: Studied project requirements, analyzed banking modules, and prepared the application structure.

Day 2: Developed customer and account classes. Implemented constructors, methods, and validation logic.

Day 3: Built the user registration module and generated unique account numbers for customers.

Day 4: Implemented account management features to display and update customer information. Corrected logical errors found during testing.

Day 5: Improved menu-driven navigation, optimized methods, reduced duplicate code, and added comments for readability.

Day 6: Tested all completed modules using different input cases, documented completed work, and discussed future enhancements with my mentor.

IV. Technical Work Completed

Implemented classes, objects, constructors, getter and setter methods, encapsulation, menu-driven programming, input validation, reusable methods, and modular code organization. Practiced arrays, strings, loops, methods, Scanner class, and exception handling. Improved coding standards by using meaningful variable names and structured programming techniques. Designed the application so that future modules such as deposits, withdrawals, and fund transfers can be integrated easily.

V. Challenges and Solutions

The biggest challenge was connecting multiple Java classes while maintaining proper program flow. I also faced runtime errors caused by invalid user input and incorrect object references. These issues were solved through systematic debugging, additional validation, mentor guidance, and repeated testing. Reading Oracle documentation and practicing similar Java programs significantly improved my understanding of professional coding practices.

VI. Skills Learned

This week strengthened my understanding of object-oriented programming, modular software development, debugging, logical thinking, code optimization, input validation, and software documentation. I also improved my communication skills by discussing implementation strategies with mentors and documenting project progress regularly.

VII. Learning Resources

Oracle Java Documentation, GeeksforGeeks, JavaTPoint, Bro Code Java tutorials, HackerRank programming exercises, mentor guidance, classroom notes, and practice programs were the primary learning resources used throughout the week.

VIII. Next Week Goals

In the upcoming week, I plan to implement deposit, withdrawal, fund transfer, transaction history, and account statement modules. I will also strengthen exception handling, perform extensive testing, optimize application performance, and continue documenting project progress.

IX. Weekly Summary

Week 03 provided valuable practical experience in applying Core Java concepts to a real-world project. I successfully developed important banking modules while improving coding standards, debugging techniques, and object-oriented design skills. The knowledge gained this week has prepared me to complete the remaining modules of the Banking Information System confidently and efficiently.